

National Coverage Determination (NCD)

Vagus Nerve Stimulation (VNS)

160.18

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Tracking Information

Publication Number

100-3

Manual Section Number

160.18

Manual Section Title

Vagus Nerve Stimulation (VNS)

Version Number

3

Effective Date of this Version

02/15/2019

Implementation Date

07/22/2020

Description Information

Benefit Category

Durable Medical Equipment

Incident to a physician's professional Service

Physicians' Services

Please Note: This may not be an exhaustive list of all applicable Medicare benefit categories for this item or service.

Item/Service Description

A. General

Vagus Nerve Stimulation (VNS) is a pulse generator, similar to a pacemaker, that is surgically implanted under the skin of the left chest and an electrical lead (wire) is connected from the generator to the left vagus nerve. Electrical signals are sent from the battery-powered generator to the vagus nerve via the lead. These signals are in turn sent to the brain. The Food and Drug Administration (FDA) approved VNS for treatment of refractory epilepsy in 1997 and for resistant depression in 2005.

Indications and Limitations of Coverage

B. Nationally Covered Indications

Effective for services performed on or after July 1, 1999, VNS is reasonable and necessary for patients with medically refractory partial onset seizures for whom surgery is not recommended or for whom surgery has failed.

Effective for services performed on or after February 15, 2019, the Centers for Medicare & Medicaid Services (CMS) will cover FDA-approved VNS devices for treatment resistant depression (TRD) through Coverage with Evidence Development (CED) when offered in a CMS-approved, double-blind, randomized, placebo-controlled trial with a follow-up duration of at least one year with the possibility of extending the study to a prospective longitudinal study when the CMS-approved, double-blind, randomized placebo-controlled trial has completed enrollment, and there are positive interim findings.

Each study must be approved by CMS and as a fully-described, written part of its protocol, must address whether VNS improves health outcomes for TRD patients compared to a control group, by answering all of the following research questions below. The details of the prospective longitudinal study must be described in the original protocol for the double-blind, randomized, placebo-controlled trial. Response is defined as a $\geq 50\%$ improvement in depressive symptoms from baseline, as measured by a guideline recommended depression scale assessment tool. Remission is defined as being below the threshold on a guideline recommended depression scale assessment tool. The following research questions must be addressed in a separate analysis for patients with bipolar and unipolar disease.

Research Questions:

- What is the rate of response (defined as person months of response/total months of study participation)?
- What is the rate of remission (defined as person months of response/total months of study participation)?
- What is the time from treatment until response scores are first achieved?
- What is the time from treatment until remission scores are first achieved?
- What are the population distributions of the maximum months of response, both consecutive and overall, separately?

- What are the population distributions of the maximum months of remission, both consecutive and overall, separately?
- What are the patient variables associated with successful treatment of TRD with VNS?
- What are the observed harms?
- What are the changes in disability, quality of life, general psychiatric status, and suicidality?

Patient Criteria:

The following criteria must be used to identify patients demonstrating TRD:

- The patient must be in a major depressive disorder (MDD) episode for $\geq$ two years or have had at least four episodes of MDD, including the current episode. In order to confirm the patient has MDD, accepted diagnostic criteria from the most current edition of the Diagnostic and Statistical Manual for Mental Disorder (DSM) and a structured clinical assessment are to be used.
- The patient's depressive illness meets a minimum criterion of four prior failed treatments of adequate dose and duration as measured by a tool designed for this purpose.
- The patient is experiencing a major depressive episode (MDE) as measured by a guideline recommended depression scale assessment tool on two visits, within a 45-day span prior to implantation of the VNS device.

Patients must maintain a stable medication regimen for at least four weeks before device implantation.

If patients with bipolar disorder are included, the condition must be carefully characterized.

Patients must not have:

- Current or lifetime history of psychotic features in any MDE;
- Current or lifetime history of schizophrenia or schizoaffective disorder;
- Current or lifetime history of any other psychotic disorder;
- Current or lifetime history of rapid cycling bipolar disorder;
- Current secondary diagnosis of delirium, dementia, amnesia, or other cognitive disorder;
- Current suicidal intent; or,
- Treatment with another investigational device or investigational drugs.

Individuals who receive placebo VNS will be offered active VNS at the end of the trial.

In addition, CMS will review studies to determine if they meet the 13 criteria listed below. If CMS determines that they meet these criteria, the study will be posted on CMS' CED website (<https://www.cms.gov/Medicare/Coverage/Coverage-with-Evidence-Development/index.html> ).

- a. The principal purpose of the study is to test whether the item or service meaningfully improves health outcomes of affected beneficiaries who are represented by the enrolled subjects.
- b. The rationale for the study is well supported by available scientific and medical evidence.
- c. The study results are not anticipated to unjustifiably duplicate existing knowledge.
- d. The study design is methodologically appropriate and the anticipated number of enrolled subjects is sufficient to answer the research question(s) being asked in the National Coverage Determination.
- e. The study is sponsored by an organization or individual capable of completing it successfully.
- f. The research study is in compliance with all applicable Federal regulations concerning the protection of human subjects found in the Code of Federal Regulations (CFR) at 45 CFR Part 46. If a study is regulated by the Food and Drug Administration (FDA), it is also in compliance with 21 CFR Parts 50 and 56. In addition, to further enhance the protection of human subjects in studies conducted under CED, the study must provide and obtain meaningful informed consent from patients regarding the risks associated with the study items and/or services, and the use and eventual disposition of the collected data.
- g. All aspects of the study are conducted according to appropriate standards of scientific integrity.
- h. The study has a written protocol that clearly demonstrates adherence to the standards listed here as Medicare requirements.
- i. The study is not designed to exclusively test toxicity or disease pathophysiology in healthy individuals. Such studies may meet this requirement only if the disease or condition being studied is life threatening as defined in 21 CFR §312.81(a) and the patient has no other viable treatment options.
- j. The clinical research studies and registries are registered on the www.ClinicalTrials.gov website by the principal sponsor/investigator prior to the enrollment of the first study subject. Registries are also registered in the Agency for Healthcare Quality (AHRQ) Registry of Patient Registries (RoPR).
- k. The research study protocol specifies the method and timing of public release of all prespecified outcomes to be measured including release of outcomes if outcomes are negative or study is terminated early. The results must be made public within 12 months of the study's primary completion date, which is the date the final subject had final data collection for the primary endpoint, even if the trial does not achieve its primary aim. The results must include number started/completed, summary results for primary and secondary outcome measures, statistical analyses, and adverse events. Final results must be reported in a publicly accessible manner; either in a peer-reviewed scientific journal (in print or on-line), in an on-line publicly accessible registry dedicated to the

dissemination of clinical trial information such as ClinicalTrials.gov, or in journals willing to publish in abbreviated format (e.g., for studies with negative or incomplete results).

- l. The study protocol must explicitly discuss beneficiary subpopulations affected by the item or service under investigation, particularly traditionally underrepresented groups in clinical studies, how the inclusion and exclusion criteria effect enrollment of these populations, and a plan for the retention and reporting of said populations in the trial. If the inclusion and exclusion criteria are expected to have a negative effect on the recruitment or retention of underrepresented populations, the protocol must discuss why these criteria are necessary.
- m. The study protocol explicitly discusses how the results are or are not expected to be generalizable to affected beneficiary subpopulations. Separate discussions in the protocol may be necessary for populations eligible for Medicare due to age, disability or Medicaid eligibility.

Consistent with section 1142 of the Act, the Agency for Healthcare Research and Quality (AHRQ) supports clinical research studies that CMS determines meet the above-listed standards and address the above-listed research questions.

The principal investigator must submit the complete study protocol, identify the relevant CMS research questions that will be addressed and cite the location of the detailed analysis plan for those questions in the protocol, plus provide a statement addressing how the study satisfies each of the standards of scientific integrity (a. through m. listed above), as well as the investigator's contact information, to the address below. The information will be reviewed, and approved studies will be identified on the CMS website.

Director, Coverage and Analysis Group
Centers for Medicare & Medicaid Services (CMS)
7500 Security Blvd., Mail Stop S3-02-01
Baltimore, MD 21244-1850

C. Nationally Non-Covered Indications

Effective for services performed on or after July 1, 1999, VNS is not reasonable and necessary for all other types of seizure disorders which are medically refractory and for whom surgery is not recommended or for whom surgery has failed.

VNS is non-covered for the treatment of TRD when furnished outside of a CMS-approved CED study.

All other indications of VNS for the treatment of depression are nationally non-covered.

D. Other

Patients implanted with a VNS device for TRD may receive a VNS device replacement if it is required due to the end of battery life, or any other device-related malfunction.

(This NCD last reviewed February 2019.)

Cross Reference

[CED page](#) 

Transmittal Information

Transmittal Number

10145

Coverage Transmittal Link

<https://www.cms.gov/files/document/R10145NCD.pdf> 

Revision History

04/2026 - Transmittal 13623 issued February 05, 2026, is being rescinded and replaced by Transmittal 13760, dated April 30, 2026, to remove business requirement 14356.4 and the corresponding spreadsheet for NCD 250.4 Treatment of Actinic Keratosis as this NCD was moved to the October 2026 CR. All other information remains the same. ([TN 13760](#) ) (CR14356)

02/2026 - The purpose of this Change Request (CR) is to provide a quarterly maintenance update of ICD-10 coding conversions and other coding updates specific to National Coverage Determinations (NCDs). No policy is being changed as a result of these updates. ([TN 13623](#) ) (CR14356)

03/2025 - The purpose of this Change Request (CR) is to provide a quarterly maintenance update of ICD-10 coding conversions and other coding updates specific to National Coverage Determinations (NCDs). No policy is being changed as a result of these updates. ([TN 13097](#) ) (CR13939)

02/2024 - The purpose of this Change Request (CR) is to provide a maintenance update of ICD-10 conversions and other coding updates specific to NCDs. ([TN 12493](#) ) (CR13507)

01/2024 - Transmittal 12350 issued November 03, 2023, is being rescinded and replaced by Transmittal 12440 dated January 3, 2024, to make changes to NCD 90.2, Next Generation Sequencing, spreadsheet to align with revisions being made to CR 13278. All other information remains the same. ([TN 12440](#) ) (CR13391)

11/2023 - The purpose of this Change Request (CR) is to provide a quarterly maintenance update of ICD-10 coding conversions and other coding updates specific to National Coverage Determinations (NCDs). No policy is being changed as a result of these updates. ([TN 12350](#)) (CR13391)

10/2023 - The purpose of this Change Request (CR) is to provide a quarterly maintenance update of ICD-10 coding conversions and other coding updates specific to National Coverage Determinations (NCDs). No policy is being changed as a result of these updates. ([TN 12319](#)) (CR13391)

10/2022 - Transmittal 11546, dated August 4, 2022, is being rescinded and replaced by Transmittal 11636, dated, October 5, 2022, to remove ICD-10 dx codes added in error to NCD 150.3, business requirement 12842.4, and restore ICD-10 dx C91.92 removed in error to NCD 110.23, business requirement 12842.3. All other information remains the same. ([TN 11636](#)) (CR12842)

08/2022 - The purpose of this Change Request (CR) is to provide a maintenance update of ICD-10 conversions and other coding updates specific to NCDs. ([TN 11546](#)) (CR12842)

06/2022 - Transmittal 11400, dated May 4, 2022, is being rescinded and replaced by Transmittal 11460, dated, June 17, 2022, to update NCD 90.2, NGS, spreadsheet to conform with changes in CR 12124, and change the implementation date for all business requirements except 12705.6 to 30 days from issuance of this correction. All other information remains the same. ([TN 11460](#)) (CR12705)

05/2022 - This Change Request (CR) constitutes a maintenance update of ICD-10 conversions and other coding updates specific to NCDs. These NCD coding changes are the result of newly available codes, coding revisions to NCDs released separately, or coding feedback received. ([TN 11400](#)) (CR12705)

04/2022 - This Change Request (CR) constitutes a maintenance update of ICD-10 conversions and other coding updates specific to NCDs. These NCD coding changes are the result of newly available codes, coding revisions to NCDs released separately, or coding feedback received. ([TN 11391](#)) (CR12606)

04/2022 - Transmittal 11264, dated February 10, 2022, is being rescinded and replaced by Transmittal 11342, dated, April 6, 2022 to (1) revise BR 12606.10 instructions for NCD 110.24, (2) BR12606.2, fix typo in NCD 160.18 spreadsheet ICD-10 G40.384, which should be G40.834, and, (3) revise implementation verbiage (no changes to the actual implementation date). All other information remains the same. ([TN 11342](#)) (CR12606)

02/2022 - This Change Request (CR) constitutes a maintenance update of ICD-10 conversions and other coding updates specific to NCDs. These NCD coding changes are the

result of newly available codes, coding revisions to NCDs released separately, or coding feedback received. ([TN 11264](#)) (CR12606)

09/2021 - This Change Request (CR) constitutes a maintenance update of ICD-10 conversions and other coding updates specific to NCDs. These NCD coding changes are the result of newly available codes, coding revisions to NCDs released separately, or coding feedback received. ([TN 11025](#)) (CR12399)

08/2021 - This Change Request (CR) constitutes a maintenance update of ICD-10 conversions and other coding updates specific to NCDs. These NCD coding changes are the result of newly available codes, coding revisions to NCDs released separately, or coding feedback received. ([TN 10963](#)) (CR12399)

01/2021 - Transmittal 10515, dated December 10, 2020, is being rescinded and replaced by Transmittal 10566, dated, January 14, 2021 to remove FISS Reason Codes (RCs) 59041, 59042, 59209, and 59210 from the spreadsheet for NCD 160.18. All other information remains the same. ([TN 10566](#)) (CR12027)

12/2020 - This Change Request (CR) constitutes a maintenance update of ICD-10 conversions and other coding updates specific to NCDs. These NCD coding changes are the result of newly available codes, coding revisions to NCDs released separately, or coding feedback received.

Previous NCD coding changes appear in ICD-10 quarterly updates that can be found at: <https://www.cms.gov/Medicare/Coverage/CoverageGenInfo/ICD10.html>, along with other CRs implementing new policy NCDs. Edits to ICD-10 and other coding updates specific to NCDs will be included in subsequent quarterly releases and individual CRs as appropriate. No policy-related changes are included with the ICD-10 quarterly updates. Any policy-related changes to NCDs continue to be implemented via the current, longstanding NCD process. ([TN 10515](#)) (CR12027)

10/2020 - This Change Request (CR) constitutes a maintenance update of ICD-10 conversions and other coding updates specific to NCDs. These NCD coding changes are the result of newly available codes, coding revisions to NCDs released separately, or coding feedback received.

Previous NCD coding changes appear in ICD-10 quarterly updates that can be found at: <https://www.cms.gov/Medicare/Coverage/CoverageGenInfo/ICD10.html>, along with other CRs implementing new policy NCDs. Edits to ICD-10 and other coding updates specific to NCDs will be included in subsequent quarterly releases and individual CRs as appropriate. No policy-related changes are included with the ICD-10 quarterly updates. Any policy-related changes to NCDs continue to be implemented via the current, longstanding NCD process. ([TN 10432](#)) (CR12027)

06/2020 - Transmittal 10145, dated May 22, 2020, is being rescinded and replaced by Transmittal 10199, dated, June 23, 2020 to update business requirement 11461.1 with clarifying language to the MACs and to extend the implementation date from June 23, 2020 to July 22, 2020. All other information remains the same. ([TN 10199](#)) (CR11461)

05/2020 - Effective for claims with dates of service on or after February 15, 2019, the Centers for Medicare & Medicaid Services covers Food and Drug Administration-approved vagus nerve stimulator devices for treatment-resistant depression through Coverage with Evidence Development when all reasonable and necessary criteria are met. This is National Coverage Determination 160.18, Vagus Nerve Stimulation. ([TN 10145](#)) (CR11461)

07/2017 - This Change Request (CR) constitutes a maintenance update of International Code of Diseases, Tenth Revision (ICD-10) conversions and other coding updates specific to National Coverage Determinations (NCDs). These NCD coding changes are the result of newly available codes, coding revisions to NCDs released separately, or coding feedback received.

Previous NCD coding changes appear in ICD-10 quarterly updates that can be found at: <https://www.cms.gov/Medicare/Coverage/CoverageGenInfo/ICD10.html>, along with other CRs implementing new policy NCDs. Edits to ICD-10 and other coding updates specific to NCDs will be included in subsequent quarterly releases and individual CRs as appropriate. No policy-related changes are included with the ICD-10 quarterly updates. Any policy-related changes to NCDs continue to be implemented via the current, long-standing NCD process. ([TN 1875](#)) (CR10184)

11/2016 - This change request (CR) is the 9th maintenance update of ICD-10 conversions and other coding updates specific to national coverage determinations (NCDs). The majority of the NCDs included are a result of feedback received from previous ICD-10 NCD CR7818, CR8109, CR8197, CR8691, CR9087, CR9252, CR9540, and CR9631. Some are the result of revisions required to other NCD-related CRs released separately.

Edits to ICD-10 and other coding updates specific to NCDs will be included in subsequent, quarterly releases as needed. No policy-related changes are included with these updates. Any policy-related changes to NCDs continue to be implemented via the current, long-standing NCD process. ([TN 1753](#)) (CR9751)

08/2016 - This change request (CR) is the 9th maintenance update of ICD-10 conversions and other coding updates specific to national coverage determinations (NCDs). The majority of the NCDs included are a result of feedback received from previous ICD-10 NCD CR7818, CR8109, CR8197, CR8691, CR9087, CR9252, CR9540, and CR9631. Some are the result of revisions required to other NCD-related CRs released separately.

Edits to ICD-10 and other coding updates specific to NCDs will be included in subsequent, quarterly releases as needed. No policy-related changes are included with these updates.

Any policy-related changes to NCDs continue to be implemented via the current, long-standing NCD process. ([TN 1708](#)) (CR9751)

04/2016 - Transmittal 1630, dated February 26, 2016, is being rescinded and replaced by Transmittal 1658 to (1) remove duplicate spreadsheet NCD210.3, (2) add missing spreadsheet NCD20.33, (3) add B/MAC to requirement 3 at request of WPS/B, (4) rename the spreadsheet titles, and, (5) provide a link to the attached spreadsheets for more efficient ease of reference and accessibility. All other information remains the same. ([TN 1658](#)) (CR9540)

12/2015 - This change request (CR) is the 3rd maintenance update of ICD-10 conversions/updates specific to national coverage determinations (NCDs). The majority of the NCDs included are a result of feedback received from previous ICD-10 NCD CR7818, CR8109, CR8197, CR8691, & CR9087. Some are the result of revisions required to other NCD-related CRs released separately that included ICD-10 coding. Implementation date: 01/04/2016 Effective date: 10/1/2015. ([TN 1580](#)) (CR9252)

08/2015 - This change request (CR) is the 3rd maintenance update of ICD-10 conversions/updates specific to national coverage determinations (NCDs). The majority of the NCDs included are a result of feedback received from previous ICD-10 NCD CR7818, CR8109, CR8197, CR8691, & CR9087. Some are the result of revisions required to other NCD-related CRs released separately that included ICD-10 coding. These updates do not expand, restrict, or alter existing coverage policy. Implementation date: 01/04/2016 Effective date: 10/1/2015. ([TN 1537](#)) (CR 9252)

05/2014 - CMS translated the information for this policy from ICD-9-CM/PCS to ICD-10-CM/PCS according to HIPAA standard medical data code set requirements and updated any necessary and related coding infrastructure. These updates do not expand, restrict, or alter existing coverage policy. Implementation date: 10/06/2014 Effective date: 10/1/2015. ([TN 1388](#)) ([TN 1388](#)) (CR 8691)

09/2012 - CMS translated the information for this policy from ICD-9-CM/PCS to ICD-10-CM/PCS according to HIPAA standard medical data code set requirements and updated any necessary and related coding infrastructure. These updates do not expand, restrict, or alter existing coverage policy. Implementation date: 01/07/2013 Effective date: 10/1/2015. ([TN 1122](#)) ([TN 1122](#)) (CR 7818)

06/2007 - CMS has determined that VNS is not reasonable and necessary for the treatment of resistant depression. Effective date 05/04/2007. ([TN 70](#)) (CR 5612)

04/1999 - Provided that procedure is safe and effective for patients with medically refractory partial onset seizures for whom surgery is not recommended or has failed. Effective date 07/01/1999. (TN 114) (CR 470)

National Coverage Analyses (NCAs)

This NCD has been or is currently being reviewed under the National Coverage Determination process. The following are existing associations with NCAs, from the National Coverage Analyses database.

- [First reconsideration for Vagus Nerve Stimulation for Treatment of Resistant Depression \(TRD\) \(CAG-00313R\)](#) 
- [Second reconsideration for Vagus Nerve Stimulation \(VNS\) for Treatment Resistant Depression \(TRD\) \(CAG-00313R2\)](#) 

Additional Information

Other Versions

Title	Version	Effective Between	
Vagus Nerve Stimulation (VNS)	3	02/15/2019 - N/A	You are here
Vagus Nerve Stimulation (VNS)	2	05/04/2007 - 02/15/2019	View
Vagus Nerve Stimulation for Treatment of Seizures	1	07/01/1999 - 05/04/2007	View